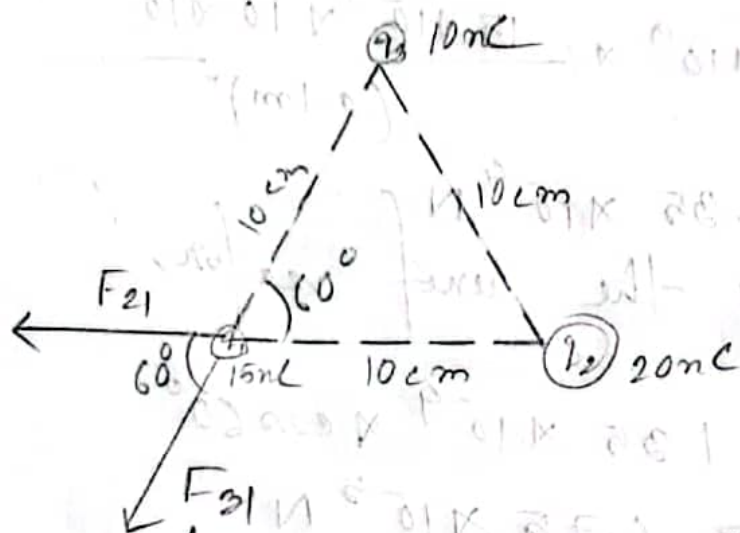


Am. to Q. 1



given that, $q_1 = 15\text{ nC} = 15 \times 10^{-9}\text{ C}$

$$q_2 = 20\text{ nC} = 20 \times 10^{-9}\text{ C}$$

$$q_3 = 10\text{ nC} = 10 \times 10^{-9}\text{ C}$$

To find the net electric force exerted on charge q_1 in unit vector and magnitude angle notation,

First,

$$\begin{aligned} F_{21} &= k_e \frac{q_1 q_2}{r^2} \\ &= 9 \times 10^9 \times \frac{15 \times 10^{-9} \times 20 \times 10^{-9}}{(0.1\text{ m})^2} \\ &= 2.7 \times 10^{-4}\text{ N} \end{aligned}$$

$$F_{31} = k_e \frac{|q_1 q_2| \cdot \text{mC}}{r^2}$$

$$= 9 \times 10^9 \times \frac{15 \times 10^{-9} \times 10 \times 10^{-9}}{(0.1 \text{ m})^2}$$

$$= 1.35 \times 10^{-4} \text{ N}$$

Now, resolving the vector
for, x-axis,

$$F_{31x} = 1.35 \times 10^{-4} \times \cos 60^\circ$$

$$= 6.75 \times 10^{-5} \text{ N}$$

y-axis,

$$F_{31y} = 1.35 \times 10^{-4} \times \sin 60^\circ$$

$$= 1.169 \times 10^{-4} \text{ N}$$

$$\therefore \vec{F}_x = (6.75 \times 10^{-5} \text{ N}) + (1.169 \times 10^{-4} \text{ N})$$

$$= 3.38 \times 10^{-4} \text{ N}$$

$$\text{and, } \vec{F}_y = 1.169 \times 10^{-4} \text{ N}$$

$$\therefore \vec{F} = (3.38 \times 10^{-4} \text{ N})\hat{i} + (1.169 \times 10^{-4} \text{ N})\hat{j}$$

$$\therefore |\vec{F}| = \sqrt{(3.38 \times 10^{-4})^2 + (1.169 \times 10^{-4})^2}$$

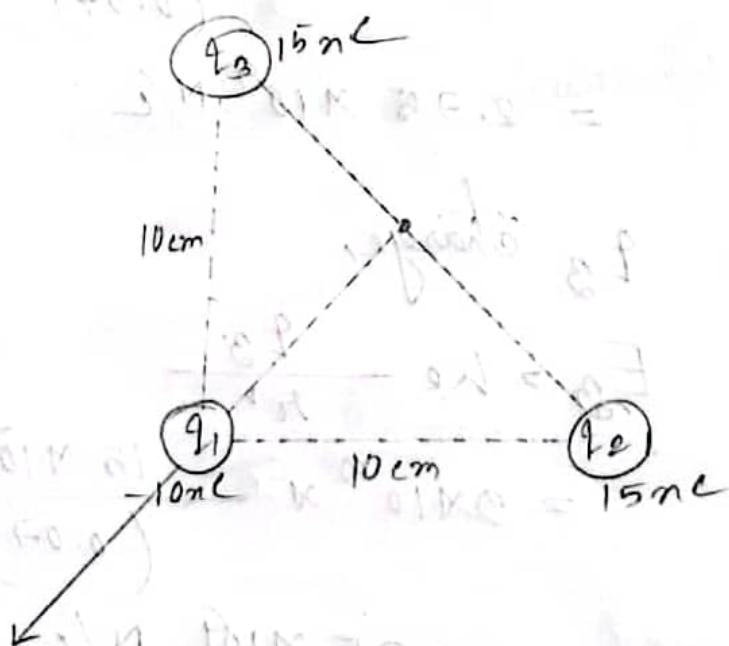
$$= 3.58 \times 10^{-4} \text{ N}$$

$$\therefore \theta = \tan^{-1} \left(\frac{1.169 \times 10^{-4}}{3.38 \times 10^{-4}} \right)$$

$$= 19.08^\circ$$

Ans

Ans. to Q. 2



given that, $q_1 = -10 \text{ nC} = -10 \times 10^{-9} \text{ C}$

$q_2 = q_3 = 15 \text{ nC} = 15 \times 10^{-9} \text{ C}$

the distance between,

$$(q_1 - q_2) = 10 \text{ cm} = 0.1 \text{ m}$$

$$(q_1 - q_3) = 10 \text{ cm} = 0.1 \text{ m}$$

$$(q_2 - q_3) = \sqrt{10^2 + 10^2} = 10\sqrt{2} \text{ cm}$$

$$= 0.141 \text{ cm}$$

Now,

To calculate the magnitude and direction of the electric field at point P,

1st, electric field for q_2 charge,

$$\begin{aligned} E_2 &= k_e \frac{q_2}{r^2} \\ &= 9 \times 10^9 \times \frac{15 \times 10^{-9}}{(0.07)^2} \quad \left| \begin{array}{l} \text{from,} \\ \text{point P to } q_2, \\ \text{distance,} \\ r = 5\sqrt{2} \text{ cm} \\ = 0.07 \text{ m} \end{array} \right. \\ &= 2.75 \times 10^4 \text{ N/C} \end{aligned}$$

again, for q_3 charge,

$$\begin{aligned} E_3 &= k_e \frac{q_3}{r^2} \\ &= 9 \times 10^9 \times \frac{15 \times 10^{-9}}{(0.07)^2} \\ &= 2.75 \times 10^4 \text{ N/C} \end{aligned}$$

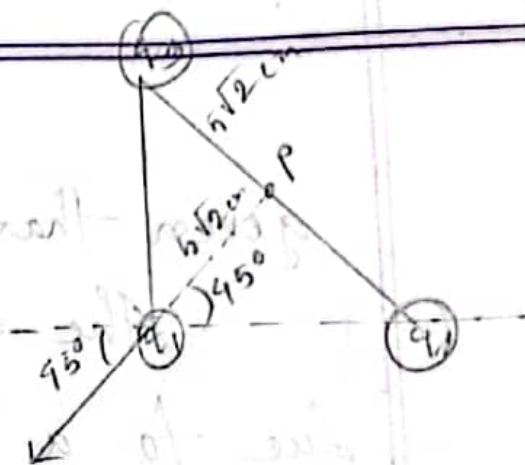
Since, $E_2 = E_3$ and their direction is opposite, both of them will cancel out each other.

Now,

$$E_1 = k_e \frac{q_i}{r^2}$$

$$= 9 \times 10^9 \times \frac{10 \times 10^{-9}}{(0.07)^2}$$

$$= 1.8 \times 10^4 \text{ N/C}$$



Since, E_2 and E_3 cancel each other,

$$E_{\text{net}} = E_1 = 1.8 \times 10^4 \text{ N/C}$$

And, the direction is towards q_1 charge at 45° .

$$\therefore \theta = 45^\circ$$

Ans

Am: 10.9.03

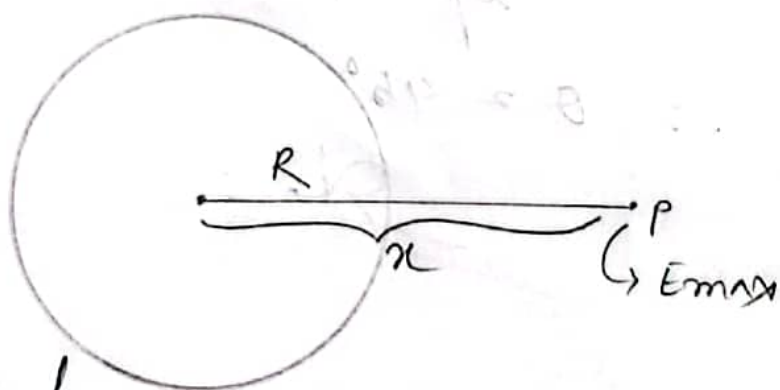
given that,

the electric field; $E = k_e x \frac{q}{(R^2 + x^2)^{3/2}}$
due to a charged ring.

R = radius of the ring.

x = distance from the center of the ring,

Let's draw the figures



At the point, P.

$$E_{\max} = k_e \frac{qx}{(R^2 + x^2)^{3/2}}$$

Electric field will be max.

$$\text{at, } \frac{dE}{dx} = 0$$

$$\therefore \frac{d}{dx} \left\{ k_e \frac{q_n}{(R^2 + x^2)^{3/2}} \right\} = 0$$

$$\Rightarrow k_e q \left[\frac{\frac{d}{dx} (R^2 + x^2)^{3/2} \cdot x - (R^2 + x^2)^{3/2} \cdot 1}{\left\{ (R^2 + x^2)^{3/2} \right\}^2} \right] = 0$$

$$\Rightarrow \frac{3}{2} (R^2 + x^2)^{1/2} \cdot 2x \cdot x - (R^2 + x^2)^{3/2} = 0$$

$$\Rightarrow 3x^2 (R^2 + x^2)^{1/2} = (R^2 + x^2)^{3/2}$$

$$\Rightarrow 3x^2 = (R^2 + x^2)$$

$$\Rightarrow 2x^2 = R^2$$

$$\therefore x = \pm R/\sqrt{2}$$

So, electric field is maximum from the center where, $x = \pm R/\sqrt{2}$.

The